HEMCHANDRACHARYA NORTH GUJARAT UNIVERSITY, PATAN
B. Sc. PHYSICS :SEMESTER -V
TYPE OF COURSE: MAJOR DISCIPLINE SPECIFIC COURSE (THEORY)

COURSE NAME: MATHEMATICAL-SOLID STATE-PLASMA PHYSICS, CLASSICAL-QUANTUM-STATISTICAL MECHANICS & ELECRONICS

PROGRAMME CODE: SCIUG103 **COURSE CODE: SC23MJDSCPHY501**

(Effective from June 2025 Under NEP–2020)

Total Credits: 04 Teaching Hours per Week: 04 Teaching Hours per Semester: 60	THEORY MAJOR	Externa lMarks (SEE) - 50 Internal Marks (CCE) - 50

Course Objective:

- To provide concepts of a differential Equations and their various coordinate systems.
- To Give basic knowledge of Mathematical Physics and to Provide deep knowledge of various Differential equation
- To provide various numerical techniques useful for the scientific data analysis.
- To Understands the concept of Langrangian Formulation and Rayleigh's Dissipation Function.
- To study the schrodinger equation, Degeneracy, Eigen value problem and Momentum eigen functions.
- To create awareness about states of Statistical Mechanics, with knowledge of Various type of ensembles.
- To provide an exposure to Drude model, Sommarfeld model, Hall co-efficient.

Course Outcome: After the successful completion of the course students will be able to,

- Understands the basic knowledge of various differential equations. various coordinate systems
- Learnt the various numerical techniques useful for the scientific data analysis.
- Solve certain special physical problems regarding mathematical aspects.
- Understands concept of Langrangian Formulation, properties of Rayleigh's Dissipation Function.
- Get sufficient knowledge of schrodinger equation, Degeneracy, Eigen value problem and Momentum eigen functions. Also knows about Physical interpretation of Eigen values, Eigenfunction.
- Create awareness about states of Statistical Mechanics like Macro and Micro states, Get Knowledge ofMicro canonical ensemble, canonical ensemble and grand canonical ensemble.
- To provide an exposure to Drude model, Sommarfeld model, Hall co-efficient.

Syllabus

UnitNo.	Content	Credit	Lect. Hrs 60
Unit-1	**MATHEMATICAL PHYSICS:** **(a) Differential Equations :** Some partial differential Equations Physics(2.1), The method of separation of variables(2.2A), Separation of Helmholtz equation in Cartesian Coordinates (2.2B), Separation of Helmholtz equation in spherical polar Coordinates(2.2C), Separation of Helmholtz equation in cylindrical coordinates(2.2D), Laplace's equation in various coordinate systems(2.2E), Choice of Coordinate system an seprability of a partial differential equation(2.3) *Related Examples, Problems & Short Questions* **(b) Second order differential Equations :** Ordinary and singular points (3.1). Series solution around and ordinary point (3.2) *Related Examples, Problems & Short Questions* **Basic Reference :** Mathematical Physics by P. K. Chatopadhyay, New age international publsihers. (Second Edition). {For b & c} **CLASSICAL MECHANICS:** **Lagrangian Formulation:** Constraints(8.1), generalized coordinates(8.2), D'alembert's principle(8.3), Lagrange's equations(8.4), A general expression for kinetic energy(8.5), Symmetries and the laws of conservation(8.6), Cyclic or ignorable coordinates(8.7), Velocity dependent potential of electromagnetic field(8.8), Rayleigh's Dissipation Function(8.9) *Related Examples, Problems & Short Questions* **Basic Reference**: Introduction to Classical Mechanics by Takawale and Puranik. McGraw Hill education (India) private limited.	1	15

Unit-II	**QUANTUM MECHANICS:** **General formalism of Wave Mechanics :** The Schrodinger equation and Probability interaction for N-particle system (3.1), The fundamental postulates of wave mechanics (3.2), The Adjoint of an operator and self Adjointness, (3.3), The Eigen value Problem; Degeneracy (3.4), Eigenvalues and Eigen Functions of Self-Adjoint Operators (3.5), The Dirac Delta Function (3.6), Observables: Completeness and Normalization of Eigen Functions (3.7), Closure (3.8), Physical Interpretation of Eigen values, Eigenfunction and expansion Co-efficients (3.9), Momentum Eigenfunctions : wave Functions in Momentum Space (3.10), Uncertainly Principle (3.11) *Related Examples, Problems & Short Questions* **Basic Reference** : A text book of Quantum Mechanics by P.M. Mathews and K. Venkateshan, McGraw Hill education (India) private limited.2nd Edition **SOLID STATE PHYSICS** **Free Electron Theory of Metals:** The Drude Model (6.1), Electrical Conductivity of Metals (6.1.1), Thermal Conductivity of Metals (6.1.2), Lorentz Modification of the Drude Model (6.2), The Fermy-Dirac (F.D.) Distribution Function (6.3), The Sommerfeld Model (6.4), Density of States(6.4.1), The Free Electron Gas at 0◦ K (6.4.2), Energy Of Electron at 0◦ K (6.4.3), The Electron Heat Capacity (6.5), The Sommerfeld Theory of Conduction in Metals (6.6), The Hall Co-efficient(RH) (6.6.1).Mathiessen's Rule (6.7). *Related Examples, Problems & Short Questions* **Basic Reference:** Elements of Solid State Physics by J.P. Srivastava, PHI New Delhi 2006 (2nd Edition)	1	15
Unit-III	**STATISTICAL MECHANICS:** **(a) Microscopic & Macroscopic States:** Microscopic states (4.1) Macroscopic States (4.2) Phase Space (4.3) μ-Space (4.4), τ-Space, (4.5) **(b) Statistical ensemble:** Micro canonical ensemble(5.1), canonical ensemble(5.2),Alternative method for the derivation of canonical distribution(5.3) Mean Value and Fluctuations(5.4), Grand canonical ensemble(5.5), Alternative derivation of Grand canonical distribution(5.6), Fluctuations in the number of particle of a system in a grand canonical distribution(5.7), Reduction of Gibb's distribution to maxwell's and Boltzman distribution(5.8), Barometric formula (5.9), Experimental verification of the Boltzman distribution(5.10) *Related Examples, Problems & Short Questions* **Basic Reference :** Fundamentalas of Statistical Mechanics 2nd edition by B B LAUD, New Age International Publishers **PLASMA PHYSICS:** **Characteristics of a Plasma in a Magnetic Field**: Description of Plasma as a Gas Mixture (3.1), Properties of Plasma in Magnetic Field (3.2), Force on Plasma in Magnetic Field (3.3), Current in Magnetised Plasma (3.4), Diffusion in a Magnetic Field (3.5), Collisions in Fully Ionized Magneto-Plasma (3.6), Pinch Effect (3.7), Oscillations and Waves in The Plasma (3.8), Plasma Frequency (3.8.1), *Related Examples, Problems & Short Questions* **Basic Reference:** Elements of Plasma Physics by S. N. Goswami New Central Book Agency (P). Ltd. Calcutta.reprint2011.	1	15
Unit-IV	**ELECTRONICS** **(a) Basic Transistor Amplifiers:** Current and Voltage amplifiers (6,10), Common Emitter Amplifiers with Emitter Resistor (6.11), Simplified Common Emitter Hybrid Model (6.12), (6.12.1, 6.12.2), Effect of An Emitter Bypass Capacitor in low frequency Response (6.13 **(b) Multistage Amplifiers :** Multistage Transistor Amplifiers (General) (7.1), Terms used in Multistage Transistor Amplifiers (7.1.1), R-C- coupled Amplifiers (7.2), Middle Frequency Range (7.2a), Low Frequency Range and Lower Cut-off Frequency (7.2b) Transformer Coupled Amplifiers (7.4, a, b), Direct coupled Amplifiers (7.5), Effect of cascading on Band width (7.6). *Related Examples, Problems & Short Questions* **Basic Reference :** Hand Book of Electronics by Gupta and Kumar. 46th revised Edition 2019	1	15

: Further Reading – Other References:

1. Mathematical Physics by B.D.Gupta.
2. Mathematical Physics by H.K.Dass.
3. Classical Mechanics, by Goldstein. Narosa Publishing House, NewDelhi.
4. Classical Mechanics by YasvantWaghmare.
5. Classical Mechanics by N. C. Rana and P. S .Joag,THM
6. Quantum Mechanics by Ghatak and Loknathan, The Macmillan Company of India Limited
7. Quantum Mechanics by Fschwabi, Narosa Publishing House, NewDelhi.
8. Quantum Mechanics by John, L. Powell and B.Crasemann.
9. Quantum Mechanics by Schiff
10. Solid State Physics by C. Kittel. John Willy and Sons.
11. Solid State Physics by Saxena. PragatiPrakashan.
12. Solid State Physics by C. M.Kachhawa.
13. Solid State Physics by S O Pillai
14. Statistical Mechanics and Properties of Matter by E.S.R. GopalMcMillan Co. of India Ltd.
15. Statistical Mechanics by B K Agarwal – Melvin Eisner, New Age Inte.Publication
16. Introduction to Plasma Physics by F.F.Chen. Plenum Press.
17. Plasma Physics by S. N. Sen, PragatiPrakashan, Meerut

HEMCHANDRACHARYA NORTH GUJARAT UNIVERSITY, PATAN

B. Sc. PHYSICS : SEMESTER – V

TYPE OF COURSE: IKS MAJOR DISCIPLINE SPECIFIC THEORY COURSE

COURSE NAME: INDIAN ASTRONOMY, ASTRO PHYSICS AND CONTRIBUTION OF INDIAN PHYSICISTS

PROGRAMME CODE: SCIUG103 **COURSE CODE: SC23MJDSCPHY501A**

(Effective from June 2025 Under NEP–2020)

Total Credits:04 Teaching Hours per Week: 04 Teaching Hours per Semester: 60	THEORY MAJOR (IKS)	External Marks - 50 Internal Marks - 50

Course Objective:

- To gain comprehensive knowledge and understanding of theoretical principles and experimental findings in Physics and its different subfields like Indian astronomy.
- To understands the Astrophysics and astronomical instruments
- To develop knowledge about magnitudes, motions and distances of star.
- To aware about Contribution of Indian Physicists

- **Course Outcome:** After the successful completion of the course students will be able to Understanding of,
- The development of astronomy from Vedic times to the recent times,The basic concepts of celestial sphere.
- The different coordinate system, Celestial longitude and latitude, Right ascension, azimuth, altitude and equinox, The Zodiac systems
- Knowledge about magnitudes, motions and distances of star.
- Contribution of Dr C V Raman, Chandrasekhar, J. C. Bose's, N. S. Satya Murthy, Homi Bhabha, Megnath Saha, S.N Bose, Dr Vikram Sarabhai, Manali Kallat Vainu Bappu.

Syllabus

UnitNo.	Content	Credit	Lect. Hrs 60
Unit-1	**INDIAN ASTRONOMY:** **Historical Introduction:** Introduction, Ancient Indian Astronomy, The Vedic Period and Vedangajyotisa, Siddhanta, Aryabhata I, Astronomers after Aryabhata, Contents of the Siddhantas, Continuity in Astronomical Tradition. **Celestial Sphere:** Introduction, Diurnal Motion of Celestial Bodies, Motion of Celestial Bodies Relative to Stars, Celestial Horizon, Meridian, Polar Star and Directions, Zodiac and Constellations, Equator and Poles, Latitude of a place and Altitude of Polar Star, Ecliptic and the Equinoxes. **Co-ordinate Systems:** Introduction, Ecliptic System, Equatorial System, Horizontal System, Meridian System, Phenomenon of Precession of Equinoxes, Ancient Indian References to the Precession, Effects of Precession on Celestial Longitude, Tropical and Sidereal Longitudes. **Rasi and Naksatra Systems:** Zodiac and Rasis. *Related Examples, Problems & Short Questions* **Basic Referenc:** 1. S. BalachandraRao, Indian astronomy: An introduction. Distributed by Orient Longman Ltd, 2000. Articles: 1.1 to 1.8, 2.1 to 2.9, 3.1 to 3.9, 4.1 2. The Story of Aastronomy in India by Chander Mohan, 2015. 3. Indian Astronomy, A Sourcebook, B. V. Subbarayappa, K. V. Sarma.	1	15
	TIME, CALENDARS AND INDIAN PANCHANGA: **Time in Indian Astronomy:** Introduction, Civil day and Sidereal day, Solar year and Civil calendar, Solar month and Lunar		

Unit-II	month, Lunar year, Adhikmasa and Ksayamasa, Yuga system, Indian eras, Time on a microcosmic scale **Calendars and Indian Panchanga:** Introduction, Gregorian calendar, Hindu calendar, Indian calendar and Panchanga, What is Panchanga, Tithi, Naksatra, Yoga, Karan, Vara **Reference Book:** 1. S. BalachandraRao, Indian astronomy: An introduction. Distributed by Orient Longman Ltd, 2000. Articles: 5.1 to 5.10, 6.1 to 6.11 2. The Story of Aastronomy in India by Chander Mohan, 2015. 3. Indian Astronomy, A Sourcebook, B. V. Subbarayappa, K. V. Sarma.	1	15
Unit-III	**ASTROPHYSICS:** **Astronomical Instruments:** Light and properties, The Earth's atmosphere and the electromagnetic radiation, Optical telescopes, Radio telescopes, The Hubble space telescope, Astronomical spectro graphs, Photographic photometry **Magnitudes, Motions and Distances of Star:** Stellar magnitude sequence, Absolute magnitude and distance modulus, The bolometric magnitude, Different magnitude standards: The UBV system and six colour photometry, Radiometric magnitudes, The colour index of a star, Luminosities of stars, Stellar parallax (Trigonometric) and the units of stellar distances, Stellar positions: The stellar coordinates, Stellar motions **Reference:** An Introduction to Astro physics by Baidyanath Basu, Tanuka Chattopadhyay and Sudhindra Nath Biswas, 2nd edition, PHI Learning private limited. Articles: 1.1 to 1.7; 3.1 to 3.10	1	15
Unit-IV	**Contribution of Indian Physicists:** **(a)** Raman's work on various musical instruments, Raman's work on magneto hydrodynamic, Chandrasekha Limit, J. C. Bose's work on electromagnetic waves, N. S. Satya Murthy's Work on Superconducting Magnets, Homi Bhabha's Work, Megnath Saha's Contribution, S.N Bose's contribution. Dr Vikram sarabhai’s contribution, Manali Kallat Vainu Bappu’s contribution and Wilson Bappu Effect. **(b)** Introduction and Activity of Various well known Indian Institute regarding Physics – PRL, ISRO, TIFR, IITs, IISc, RRI, SINP, IASC, IISST- Thiruvananthpuram, IPR, IUCAA 1. Raman and his effect by G. Venkataraman, University press. 2. Chandrasekhar and his limit by G. Venkataraman, University press Reference: Internet Archives on relevant topics. 3. Visit Official website of concern institute for reference {**For (b)**}	1	15